Multiple-choice questions

|!|BCEMB0310000001|!| ★

Directions: The following two questions refer to the passage below.

> Charles Darwin studied 13 different types of finches on the Galapagos Islands. The different types of finches had beaks of different shapes for feeding on different types of food. He suggested that seed-eating finches had first arrived on the Galapagos Islands from the mainland of South America. However, seeds available on the islands were not sufficient to feed all the finches, so the population of the seed-eating finches dropped. Some of the finches having beaks of different shapes and size could feed on other types of food such as insects. These finches could survive and reproduce. The population of insect-eating finches thus increased over many generations.

Which of the following statements about the finches is/are correct?

(1) Genetic variations exist among the finches in the shapes and size of their beaks.
(2) There was competition among the finches for food resources on the Galapagos Islands.
(3) The seed-eating finches were all eliminated.

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

□

##B##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0310000002|!| ★

The insect-eating finches could survive and reproduce while many seed-eating finches died. This was because

(1) insects contain more carbohydrates and proteins than seeds.
(2) less competition existed among the finches feeding on insects.
(3) much more seed-eating finches were eaten by their predators.

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0310000003|!|

Which of the following was the scientist who proposed the theory of natural selection to explain evolution?

A. Gregor Mendel
B. Jean Baptiste Lamarck
C. Charles Darwin
D. Carolus Linnaeus

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0310000004|!|

Which of the following is/are key factor(s) of the theory of natural selection proposed by Charles Darwin?

(1) Variations exist among individuals of the same species such that some individuals are better adapted to the environment than others.
(2) Organisms with variations favourable to their environment are able to survive and reproduce, passing their favourable characters to the offspring.
(3) Over-production of offspring and limited resources cause competition within the species.

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A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

☐

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0310000005|!| ★

According to the theory of natural selection, evolution requires
A. production of a large number of offspring.
B. isolation of populations.
C. behavioural changes.
D. variations in heritable characters.

☐

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0310000006|!| ★

Which of the following statements about natural selection is ***incorrect***?
A. Natural selection acts on a population and eventually the population becomes better adapted to its environment.
B. Natural selection creates new characteristics in organisms to help them adapt to the environment.
C. Changing conditions in the environment favour a particular form of the species to survive and reproduce more successfully.
D. Within a population, the proportion of the favoured form increases over generations.

☐

##B##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0310000007|!|

Which of the following correctly explain(s) the diversified beak shapes in finches of the Galapagos Islands?

(1) Food resources on the islands are abundant and the finches can choose which type of food to feed on.
(2) Different types of food are available on the different islands.
(3) Finches with beaks well adapted for the type of food available can find sufficient food and they have greater chance to survive and reproduce.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0310000008|!|

Which of the following statements correctly explain(s) the evolution of giraffes according to Darwin's theory of natural selection?

(1) The environment favoured those giraffes with longer necks and they had better chance of survival.
(2) Shortage of grasses caused the giraffes to lengthen their necks to feed on leaves higher up on trees.
(3) The acquired characters of long necks could be passed on to offspring.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

##A##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0310000009|!|

Which of the following is the ***best*** explanation for the fact that most giraffes have long necks today?

A. Giraffes grew longer necks during their lifetime to reach leaves high up in trees and the long neck character was passed on to their offspring.
B. Giraffes that were born with longer necks than others could survive and reproduce better as they were better adapted to the environment.
C. The giraffes with short necks were more easily preyed upon by predators.
D. Fossils of short necked giraffes and long necked giraffes have been discovered.

□

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0310000010|!| ★

Both Charles Darwin and Lamarck proposed their theories of evolution. What was the similarity between their theories?

A. Both theories mentioned that variations existed within species.
B. Both theories stated that individuals could develop new characters to adapt to the environment.
C. Both theories stated that parents were able to alter the characteristics of their offspring.
D. Both theories stated that the characters of individuals could be passed from parents to offspring.

□

##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0310000011|!| ★

Which of the following comparisons between Darwin's and Lamarck's theory of evolution is ***incorrect***?

	Darwin's theory	***Lamarck's theory***
A.	Individuals are born with characteristics that are slightly different from others.	Individuals develop new characteristics during their lifetime.
B.	Environmental changes favour the individuals having favourable characteristics to survive and reproduce.	Environmental changes cause individuals to change their characteristics to adapt to the environment.
C.	Only favourable characteristics are passed on from parents to offspring.	Characteristics acquired during lifetime can be passed on to offspring.
D.	The proportion of individuals without the selective advantage in a population would gradually decrease.	Structures that are seldom used would gradually weaken and become smaller.

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0310000012|!| ★

Initially, nearly all the peppered moths were white in Britain. After the Industrial Revolution, the number of white peppered moths decreased while the number of black peppered moths increased.

Which of the following was the main selection pressure in the above example?

A. competition among moths
B. predation by birds
C. air pollution
D. all of the above

##B##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0310000013|!| ★

Which of the following statements correctly explain(s) the changes in the peppered moth population with the theory of natural selection?

(1) The colour of the peppered moth is an inheritable trait.
(2) The white peppered moths turned black due to heavy air pollution after the Industrial Revolution.
(3) In an environment with heavy air pollution, the dark colour of the black peppered moths is favourable for survival and is selected for.

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

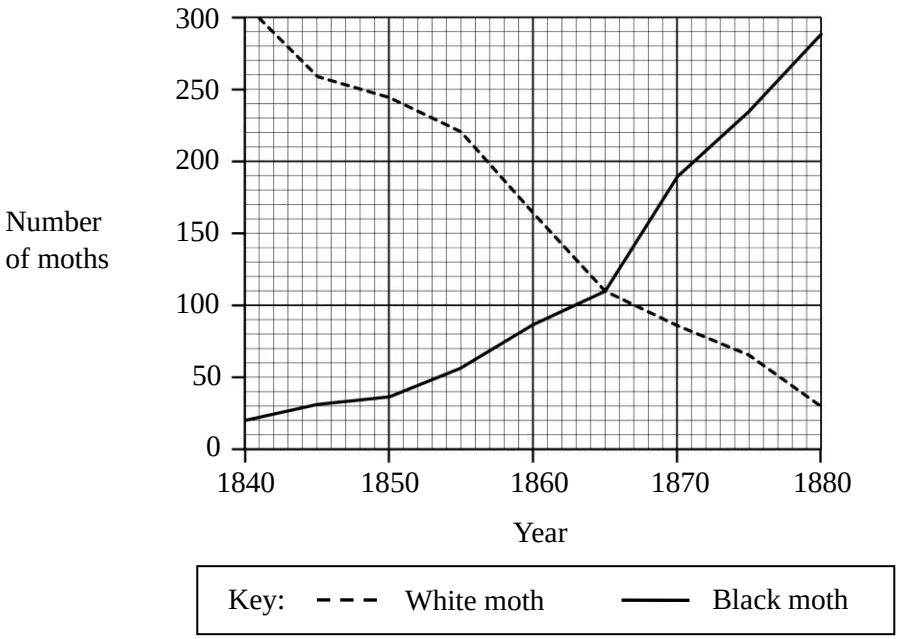
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##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0310000014|!| ★

Directions: The following three questions refer to the graph below which shows the changes in the populations of white peppered moths and black peppered moths in a woodland in England from 1840 to 1880. The tree bark was originally light green in colour. From 1840 onwards, air pollution caused by the Industrial Revolution led to blackening of the tree bark.



Determine the proportion of white peppered moths in the woodland in 1865 from the graph.

A. 0.02
B. 0.1
C. 0.3
D. 0.5

##D##

Skill assessed	Mark
Mathematical skills	1

|!|BCEMB0310000015|!| ★

What conclusion can be drawn from the graph?

A. The number of black moths was directly proportional to time.
B. The rate of increase in the black moth population was approximately equal to the rate of decrease in the white moth population.
C. The peppered moth population would be eliminated by 1890.
D. Air pollution was the direct cause for the decrease of the white moth population.

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0310000016|!| ★

The Clean Air Act was passed in 1950 in England. Air pollution was greatly reduced afterwards. Which of the following is the ***most probable*** reason for the change in the proportion of white and black moths in the 50 years after the Clean Air Act?

A. The proportion of white moths increased because they were less readily seen by birds on light-coloured tree trunks.
B. The proportion of white moths decreased because they were killed by a new disease.
C. The proportion of black moths increased because more black moths moved into the woodland from other areas.
D. The proportion of black moths decreased because more lichens and mosses could grow on tree trunks and they compete with black moths for resources.

##A##

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Skill assessed	Mark
Applying concepts	1

|!|BCEMC0310000017|!|

Which of the following can lead to the development of antibiotic resistance in bacteria?

(1) mutation of bacteria
(2) widespread use of vaccines
(3) patients stop taking antibiotics once disease symptoms disappeared

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0310000018|!|

The indiscriminate use of antibiotics promotes the emergence of antibiotic-resistant bacteria. This means that

(1) bacteria acquire resistance against antibiotics upon exposure to the antibiotics.
(2) the antibiotic-resistant bacteria are selected for and their proportion in the population increases.
(3) the non-resistant bacteria are selected against and their proportion in the population decreases.

A. (1) only
B. (2) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##C##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0310000019|!|

Nowadays some antibiotics have become ineffective against bacteria. Which of the following statements correctly explain(s) the rapid evolution of antibiotic-resistant bacteria?

(1) Antibiotic-resistant bacteria reproduce faster than the non-resistant ones.
(2) Antibiotic-resistant bacteria have a better chance to survive and reproduce than the non-resistant ones.
(3) Bacteria usually have short generation time.

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0310000020|!| ★

In the 20th century, DDT had been widely used for mosquito control but its effectiveness against mosquitoes had been decreasing. This was because

A. the use of DDT was banned.
B. DDT caused mosquitoes to develop resistance against DDT.
C. the reproduction rate of mosquitoes increased after exposure to DDT.
D. DDT-resistant mosquitoes survived and reproduced.

##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0310000021|!|

The use of pesticides to kill pests was effective initially but upon repeated use the effectiveness might decrease. Which of the following statements about the emergence of pesticide-resistant pests are correct?

(1) There are genetic variations among pests in their resistance against pesticides.
(2) The pests resistant to pesticides pass the resistance character to their offspring.
(3) The pests resistant to pesticides kill the non-resistant pests.

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##B##

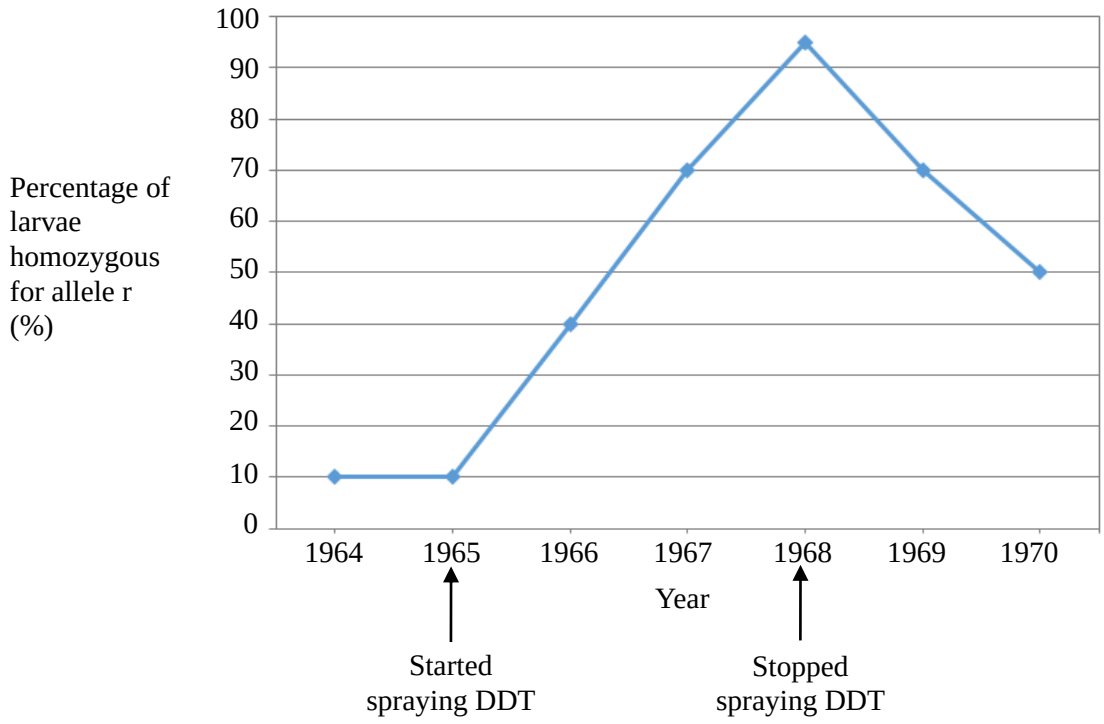
Skill assessed	Mark
Applying concepts	1

|!|BCEMB0310000022|!| ★

In the mid-1960s, DDT was widely used as an insecticide against mosquitoes. The sensitivity to DDT in mosquitoes is determined by a single gene that has two alleles:

- allele r: resistant to DDT
- allele R: sensitive to DDT

The prevalence of allele r in a population of mosquito larvae was measured from 1964 to 1970. The graph below shows the results.



Which of the following conclusion can be drawn from the graph?

A. Mosquitoes which are homozygous for allele R was completely eliminated from the population in 1968.
B. Mosquitoes which are homozygous for allele r would be completely eliminated from the population after 1970.
C. During the widespread use of DDT, mosquitoes which are homozygous for allele r had a greater chance of survival than the mosquitoes which are homozygous for allele R.
D. After the use of DDT was banned in 1968, mosquitoes which are homozygous for allele R had selective advantage over the mosquitoes which are homozygous for allele r.

□

##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMC0310000023|!|

Which of the following statements about sickle-cell anaemia is/are correct?

(1) Sickle-cell anaemia is a genetic disorder caused by a substitution of one base in the gene coding for a polypeptide chain of haemoglobin.
(2) Heterozygous carriers have a selective advantage over people with normal red blood cells in areas where malaria is common.
(3) In areas where malaria is common, heterozygous carriers and people with normal red blood cells have a greater chance of survival than people homozygous for the sickle-cell allele.

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0310000024|!|

When compared with people with normal haemoglobin, carriers of sickle cell anaemia are more resistant to

A. AIDS.
B. influenza.
C. anaemia.
D. malaria.

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0310000025|!|

A new species is considered to have evolved when different groups of a population

A. are isolated from each other by a geographic barrier.
B. show increased variations due to mutations.
C. can no longer interbreed to produce fertile offspring.
D. are subjected to increased competition due to limited resources in their habitats.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0310000026|!|

Which of the following is ***not*** a requirement for speciation to occur?

A. genetic variations in a population
B. gene flow
C. isolation
D. changes in environmental conditions

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0310000027|!|

Which of the following descriptions about speciation is ***incorrect***?

A. For speciation to occur, a population must be separated into isolated groups.
B. Speciation does not occur if individuals of a population can freely interbreed and produce fertile offspring.
C. Under the same selection pressure, different favourable variations may be selected by natural selection in each isolated group.
D. The isolated groups are said to have become different species when members of each group reach each other again but can no longer interbreed to produce fertile offspring.

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0310000028|!|

Which of the following conditions is/are essential for the formation of new species?

(1) mutation
(2) natural selection
(3) gene flow between populations

A. (2) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0310000029|!|

Which of the following organisms are ***less*** likely to be geographically isolated by mountain ranges?

A. mammals
B. fish
C. flowering plants
D. birds

##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0310000030|!|

Which of the following statements about isolation is/are correct?

(1) Isolation stops gene flow between the isolated groups of a population.
(2) Geographic isolation refers to separation of a population by physical barriers such as oceans and mountain ranges.
(3) Isolation leads to increased rate of mutation in the isolated groups.

A. (1) only
B. (2) only
C. (1) and (2) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Applying concepts	1